**5** **(a)** As the triangular loop exits the field, the area of the loop remaining inside the field decreases.

The magnetic flux through the loop decreases and by Faraday's law, an e.m.f. is induced in the loop.

**(b)** Once the triangle starts to exit the field, the area within the magnetic field decreases by a triangular segment where the base $(vt)$ and height (which is the same as the base) are both proportional to time.

$$\therefore A = A_0 - \frac{1}{2}(vt)(vt) = A_0 - \frac{1}{2}v^2t^2$$

$$E = \frac{d(BA)}{dt} = \frac{Bd\left(A_0 - \frac{1}{2}v^2t^2\right)}{dt}$$

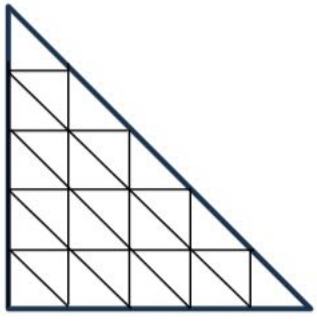
$$= Bv^2t$$

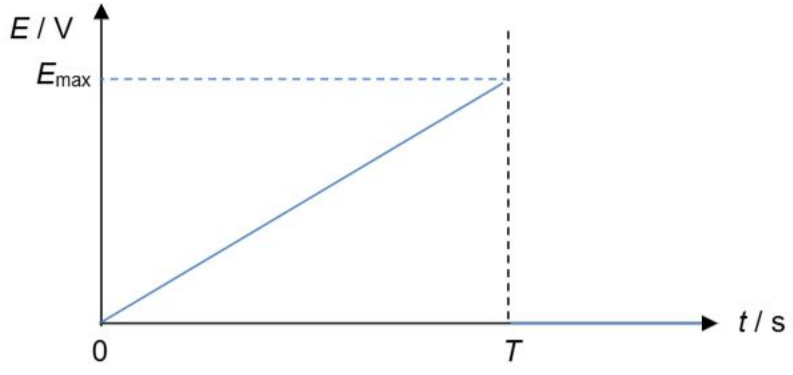
$$\Rightarrow E \propto \frac{dA}{dt} \propto t$$

OR

In the first $\Delta t$, area decreases by 1 small triangle.
In the second $\Delta t$, area decreases by 3 small triangles.
In the third $\Delta t$, area decreases by 5 small triangles.
In the fourth $\Delta t$, area decreases by 7 small triangles.

Hence, the rate of decrease of area increases at a constant rate and $E$ increases at a constant rate.






*$E$ increasing linearly until T

**(c)** Total change in area $= \frac{1}{2}(0.15)^2 = 0.01125 \text{ m}^2$

Total time taken $= \frac{0.15}{0.30} = 0.50 \text{ s}$

$$E_{ave} = \frac{\Delta\phi}{\Delta t} = \frac{B\Delta A}{\Delta t} = \frac{(0.40)(0.01125)}{0.50}$$

$$= 0.0090 \text{ V}$$